

# Year 10 Mathematics

## Composite Area, Surface Area and Volume

### Resource Free Section

**Instructions:** Answer all questions. Formula sheet required.

Time: 10 mins Name: Marking Key /10 marks

1. Find the area of a 4m by 4m square. (2 mark)

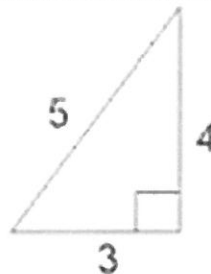
$$4 \times 4 = 16 \checkmark \quad 16m^2 \checkmark$$

2. Determine the area of the triangle if the units are in metres. (2 marks)

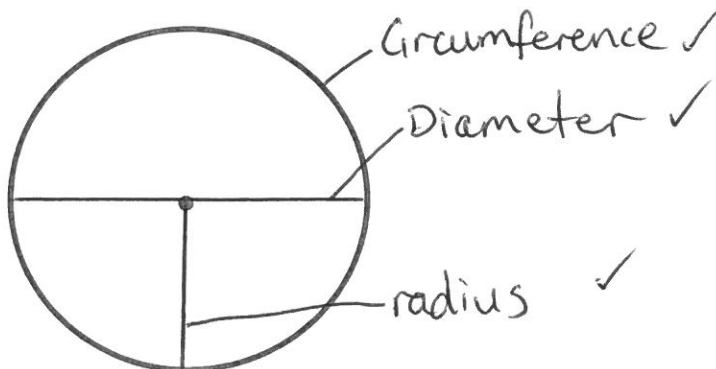
$$A = \frac{1}{2}bh$$

$$A = \frac{1}{2} \times 3 \times 4 \checkmark$$

$$= 6m^2 \checkmark$$



3. Label the circle with *radius*, *diameter*, and *circumference*. (3 marks)



4. Determine the volume of the prism with a square end area of  $16cm^2$  and the other dimension is 10cm long. (2 marks)

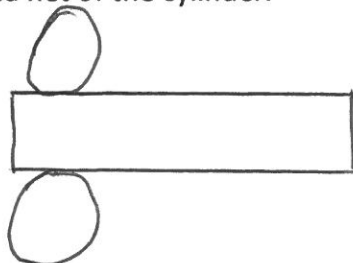
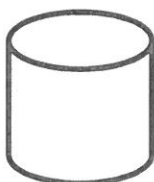


$$V = A \times h$$

$$V = 16 \times 10 \checkmark$$

$$V = 160cm^3 \checkmark$$

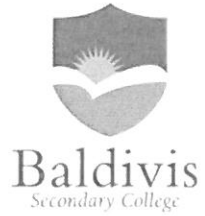
5. Draw a surface area net of the cylinder. (1 mark)



# Year 10 Mathematics

## Composite Area, Surface Area and Volume

### Resource Section



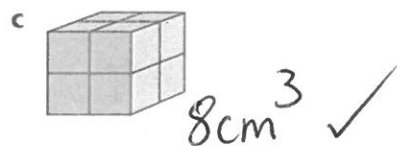
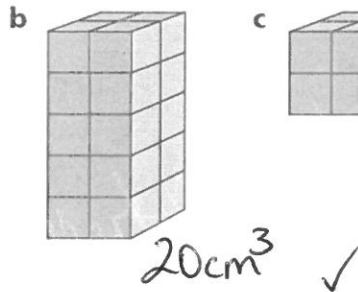
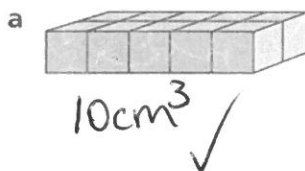
**Instructions:** Answer all questions and show all working.

**Resources:** 1 side of A4 notes are allowed, scientific non-programmable calculator allowed.

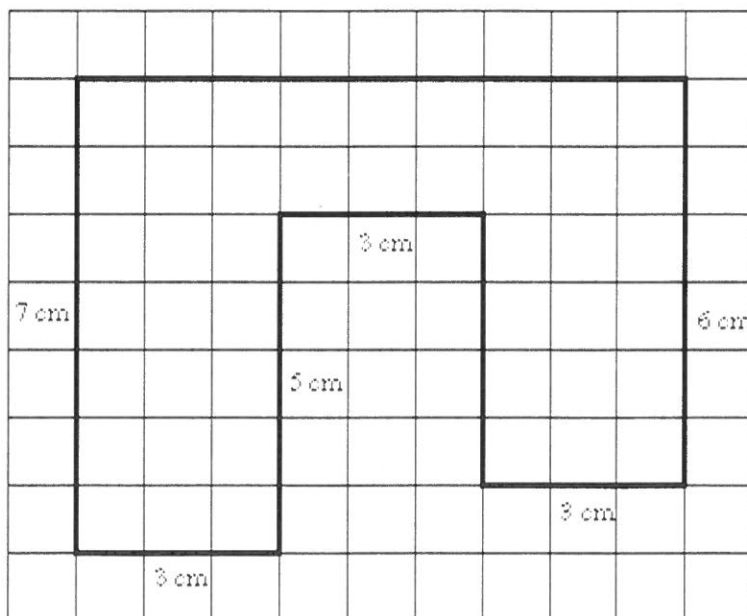
Time: 45 mins Name: Marking Key. /35 marks

Total /45 marks

6. Determine the volume of the following prisms if each cube is 1 cubic cm. (3marks)



7. Determine the area of the compound shape. (2 marks)

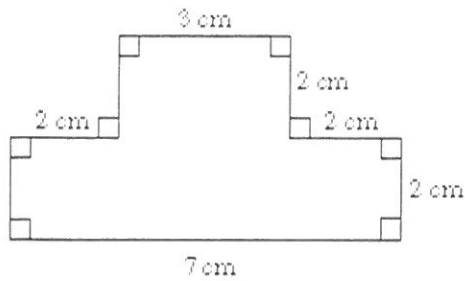


$$\begin{aligned} 6 \times 3 &= 18 \checkmark \\ 3 \times 3 &= 9 \checkmark \\ 7 \times 3 &= 21 \checkmark \end{aligned}$$

$$18 + 9 + 21 = 48\text{cm}^2 \checkmark$$

8. Calculate the perimeter and area of the compound shape.

(5 marks)



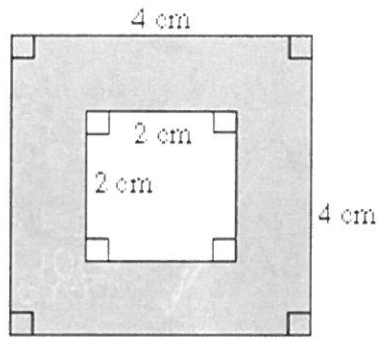
$$\begin{aligned} \text{Area} &= 3 \times 2 + 7 \times 2 \\ &= 20 \text{ cm}^2 \end{aligned}$$

$$\text{Perimeter} = 3 + 2 + 2 + 2 + 7 + 2 + 2 + 2 = 22 \text{ cm}$$

9. Determine the size of the shaded area (show all working)

(3, 3 = 6 marks)

a.

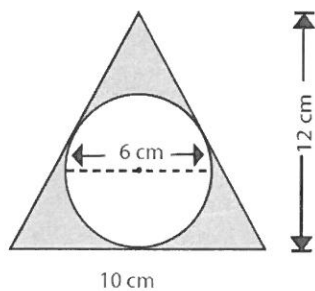


$$\begin{aligned} A &= 4 \times 4 \\ &= 16 \text{ cm}^2 \end{aligned}$$

$$16 - 4 = 12 \text{ cm}^2$$

$$\begin{aligned} A &= 2 \times 2 \\ &= 4 \text{ cm}^2 \end{aligned}$$

b.



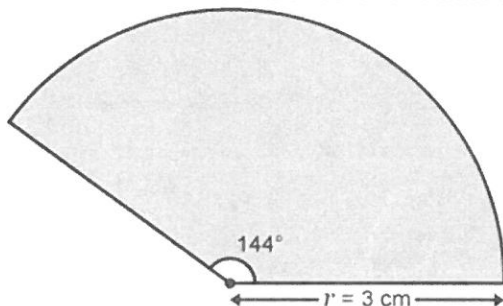
$$\begin{aligned} A &= \frac{1}{2} \times 10 \times 12 \\ &= 60 \text{ cm}^2 \end{aligned}$$

$$60 - 28.27 = 31.73 \text{ cm}^2$$

$$\begin{aligned} A &= \pi \times 3^2 \\ &= 28.27 \text{ cm}^2 \end{aligned}$$

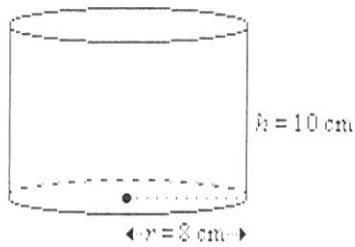
10. Determine the area of the sector.

(2 marks)

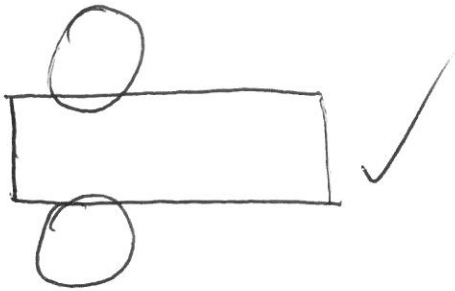


$$\frac{144}{360} \times \pi \times 3^2 = 11.31 \text{ cm}^2$$

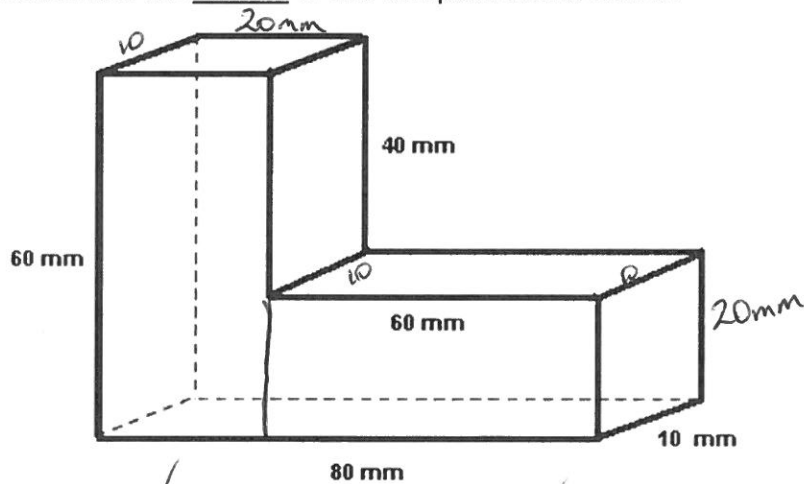
11. Calculate the surface area of the cylinder. Round your answer to the nearest whole square centimetre. (Use a drawing to help you) (5 marks)



$$\begin{aligned} \text{TSA} &= 2 \times \pi \times 8^2 + 2 \times \pi \times 8 \times 10 \\ &= 904.778 \text{ cm}^2 \\ \text{Rounded } &905 \text{ cm}^2 \end{aligned}$$



12. Determine the volume of the composite solid below. (3 marks)

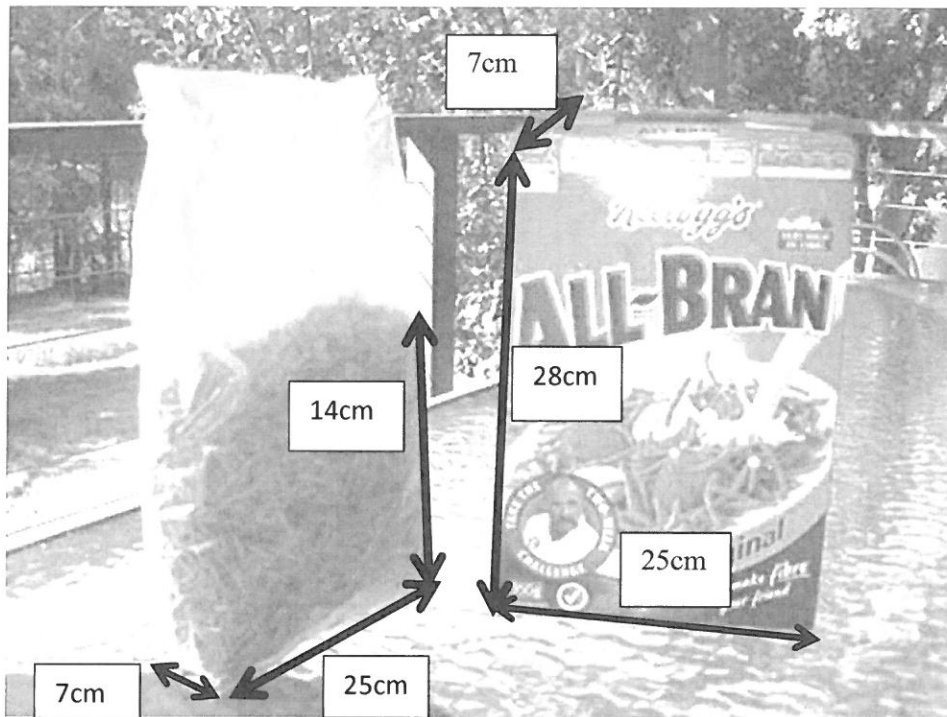


$$\begin{aligned} V &= 60 \times 10 \times 20 + 60 \times 10 \times 20 \\ &= 24000 \text{ mm}^3 \end{aligned}$$

13.

$$\text{Box Volume} = 25 \times 28 \times 7 \\ = 4900 \text{ cm}^3 \checkmark$$

$$\text{Cereal Volume} = 7 \times 25 \times 14 \\ = 2450 \text{ cm}^3 \checkmark$$



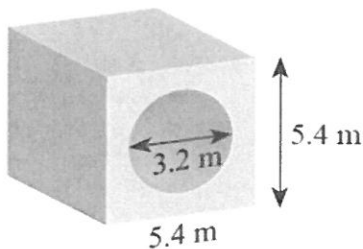
$$4900 - 2450 \\ = 2450 \text{ cm}^3 \checkmark$$

The amount not occupied is  $2450 \text{ cm}^3$  ✓

From the pictures above calculate the amount of space inside the box not occupied by cereal. Full workings required. (4 marks)

14. Determine the surface area and volume of the solid below.

(5marks)



$$\text{TSA} = \left[ (5.4^2 \times 6) - (2 \times \pi \times 1.6^2) \right] + \\ (2 \times \pi \times 1.6 \times 5.4) \\ = 213.16 \text{ m}^2$$

$$\text{Volume} = (5.4^3) - (\pi \times 1.6^2 \times 5.4) = 114.03 \text{ m}^3$$